EECS 182
Final Project Winter 2026

Due Wednsday March 17, 2026

Select a suitable device among the 3 provided in the data sheet to design a single stage amplifier to meet the following requirements

Gain (G_A) > 18 dB
Noise figure (NF) ≤ 1.7 dB
Frequency 1 GHz
$VSWR_{INPUT}$ < 3:1 (*Note this spec is critical and must be meet*)
$VSWR_{OUTPUT}$ 1:1 1 (*Note this spec is critical and must be meet*)
$(\text{Stability margin})_{INPUT}$ ≥ 0.05
$(\text{Stability margin})_{OUTPUT}$ ≥ 0.05

To complete the project follow the steps below:

1) Analyze the requirements and describe your approach for the design of the amplifier.

2) For each device check stability, if the device is not unconditionally stable draw the stability circles at the input and output on two separate Smith's charts for the input and output plane.

3) For each device draw the available gain circles in the proper plane.

4) For each device draw the noise figure circles in the proper plane.

5) Select the device that satisfy or exceed the requirements specified above, explain your choice and explain why you select one device vs. the other two.

6) For the selected device tune Γ_s to trade off between Gain, noise figure, stability margin and $(VSWR)_{INPUT./OUTPUT}$ Note this last specification **<u>must be meat at all costs</u>**.

7) Select the optimum value for the input Γ_s accordingly to the amplifier specifications and stability requirements. Pay particular attention to the $VSWR_{IN}$ requirement and make sure you meet this specification.

8) Select the correspondent value of Γ_L and verify the stability requirements for it in the proper plane.

9) Use two separate Smith's charts to design the input and output, matching networks. Use balanced open stubs having characteristic impedance of 100Ω in combination with transmission line (as in Fig.1) to design the matching circuit.

10) Realize your matching circuits in MIC technology using microstrip lines for the matching stub use open stubs having impedance of 100 Ω. (use the graphs in Fig.3 to design all the microstrip lines). Use Rogers as substrate (ε_r=4, h=0.254 mm) and 201 form factor for the passive components (see Fig. 2) for the bias circuit and the decoupling network.

11) Design the bias network for the device including the decupling capacitors as follow:

- Use High/Low impedance (200 Ω /20Ω) quarter wave line/stub configuration to provide the bias to the gate and drain of the device, the source should be connected to ground trough a via hole.
- Assume differential voltage Vcc = +5V/-5V supply is available for the bias of the device.
- You can assume no current on the gate (I_G~0) of the device when biasing the gate (V_{GS}), determine the value of the resistors R_1 and R_2.
- Use the I-V curve in the data sheet and draw the proper load line to calculate the resistors R_3 needed to bias the drain (V_{DS}).
- In addition to their value also compute the power dissipated by each resistor used in the bias network.
- Compute the values for the choke inductors L_{CK} and decoupling capacitors C_{DEC}.

12) Provide a good draw to scale of the circuit layout (make sure it is "to scale") including matching network, bias network and decoupling capacitors.

13) Summarize your final amplifier circuit and performance trade off.

14) If your design cannot meet all the requirements explain why, discuss what you learn during this project and explain your tradeoffs.

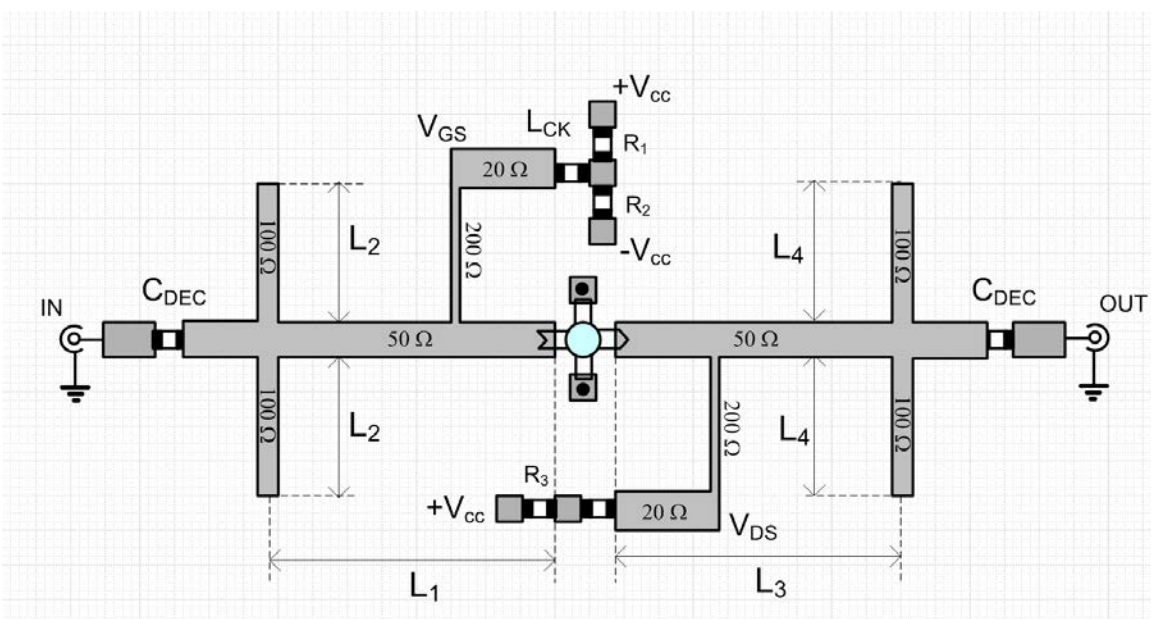

Fig.1 Amplifier final layout (not too scale)

Code		Length (l)		Width (w)		Height (h)	
Imperial	Metric	inch	mm	inch	mm	inch	mm
0201	0603	0.024	0.6	0.012	0.3	0.01	0.25
0402	1005	0.04	1.0	0.02	0.5	0.014	0.35
0603	1608	0.06	1.55	0.03	0.85	0.018	0.45
0805	2012	0.08	2.0	0.05	1.2	0.018	0.45
1206	3216	0.12	3.2	0.06	1.6	0.022	0.55
1210	3225	0.12	3.2	0.10	2.5	0.022	0.55
1812	3246	0.12	3.2	0.18	4.6	0.022	0.55
2010	5025	0.20	5.0	0.10	2.5	0.024	0.6
2512	6332	0.25	6.3	0.12	3.2	0.024	0.6

Fig.2 SMD component layout and dimensions (use 0201)

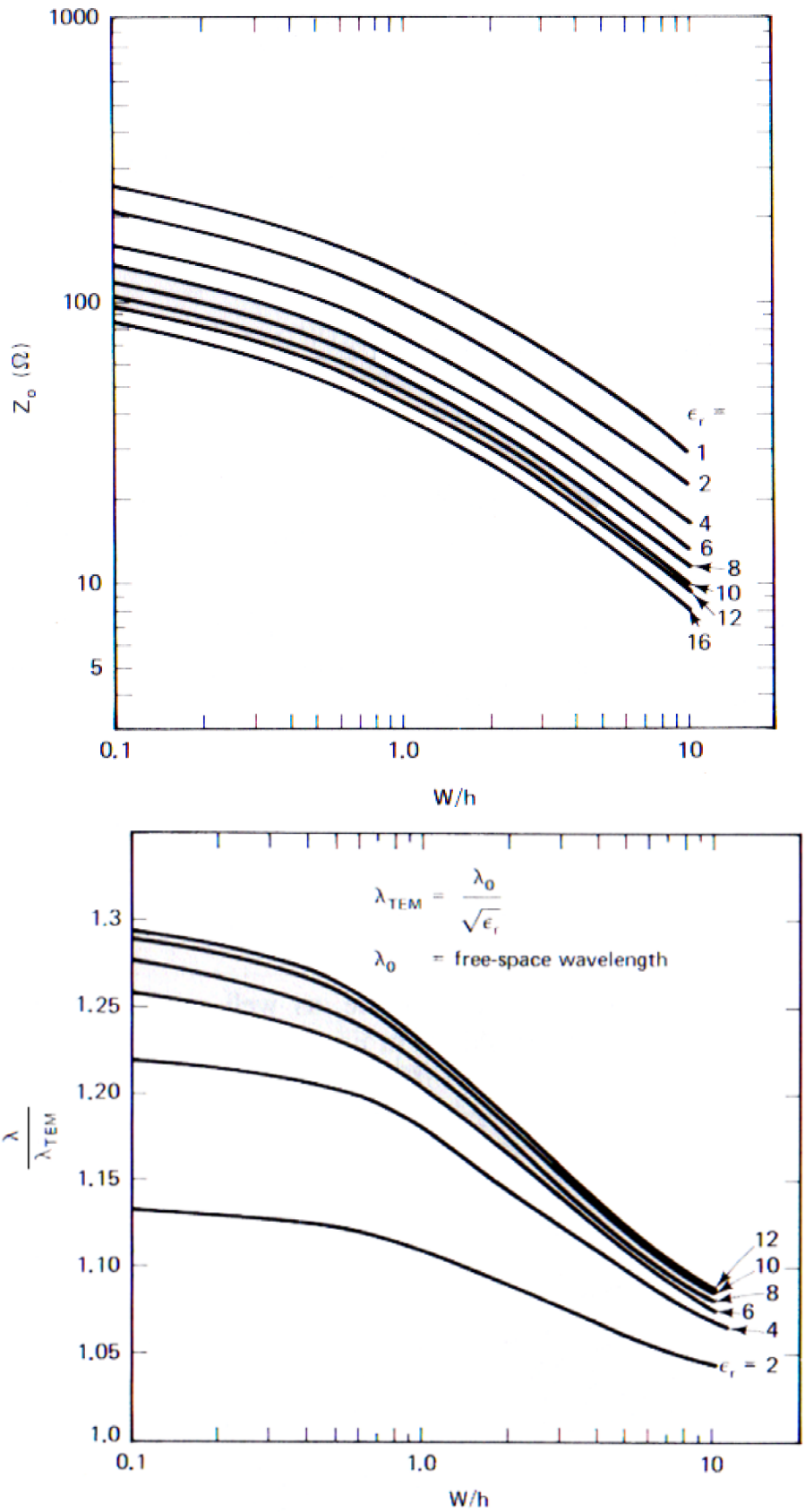

Fig.3 Microstrip design curves